

Dross Vault dMRV: Digital MRV for CBG Plants

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Summary

Dross Vault dMRV is a digital measurement, reporting and verification system for compressed biogas (CBG) plants. It replaces manual logbooks with sensor data, calculates the plant's carbon credits automatically using an approved methodology, makes every record tamper-evident, and tracks each credit from issuance to retirement in an evidence-linked registry.

A CBG plant earns credits for two things: methane avoided by processing organic waste that would otherwise rot or be burned, and fossil fuel displaced by the CBG it sells. Today the data behind those credits comes from weighbridge slips, logbooks and periodic site audits, which makes verification slow, costly and open to error or manipulation.

The system has five layers:

1. **Measurement:** weighbridge, gas flow meters, gas analyzers, methane leak detectors and energy meters at each stage.
2. **Data pipeline:** an edge gateway that timestamps, signs and streams readings to the cloud.
3. **Calculation engine:** the crediting methodology written as code, computing baseline emissions, project emissions and net reductions.
4. **Integrity layer:** validation rules plus a hash-chained ledger that makes any later edit detectable.
5. **Reporting and registry:** live dashboards, auto-generated monitoring reports, a read-only auditor view, and a registry that links each credit to its source data.

The pilot is the Dross Vault Guwahati plant; the market is 132 commissioned CBG plants and over 1,100 more holding SATAT letters of intent.

Target market and plant conditions

The target market is India's CBG plants under the SATAT scheme: 132 plants commissioned with 920 tonnes per day of capacity ([IamRenew](#)), over 1,100 letters of intent, and a 5,000-plant target ([MarkNtel Advisors](#)). Every one of these plants can earn carbon credits, and every one needs trusted data to do so.

Why CBG plants need dMRV.

Problem today	What dMRV changes
Feedstock tonnage and type recorded by hand	Weighbridge readings logged automatically, by feedstock type
Gas output estimated from compressor logs or invoices	Continuous flow and methane measurement on raw biogas, off-gas and product
Methane slip and leaks assumed or ignored	Measured, so project emissions are accurate and defensible
Verifier must visit and reconcile paper	Remote read-only access with re-runnable integrity checks
Credits issued months after the reductions	Monitoring reports generated as soon as a period closes
Buyers doubt credit quality	Every credit traceable to the sensor records behind it

Design choices for plant conditions. Remote rural sites with patchy internet get an edge gateway that buffers data offline. Mixed-vendor equipment is handled with non-intrusive meters and read-only Modbus connections. Small technical teams get simple dashboards and alerts by SMS or WhatsApp. Records kept in spreadsheets or Tally are imported rather than replaced.

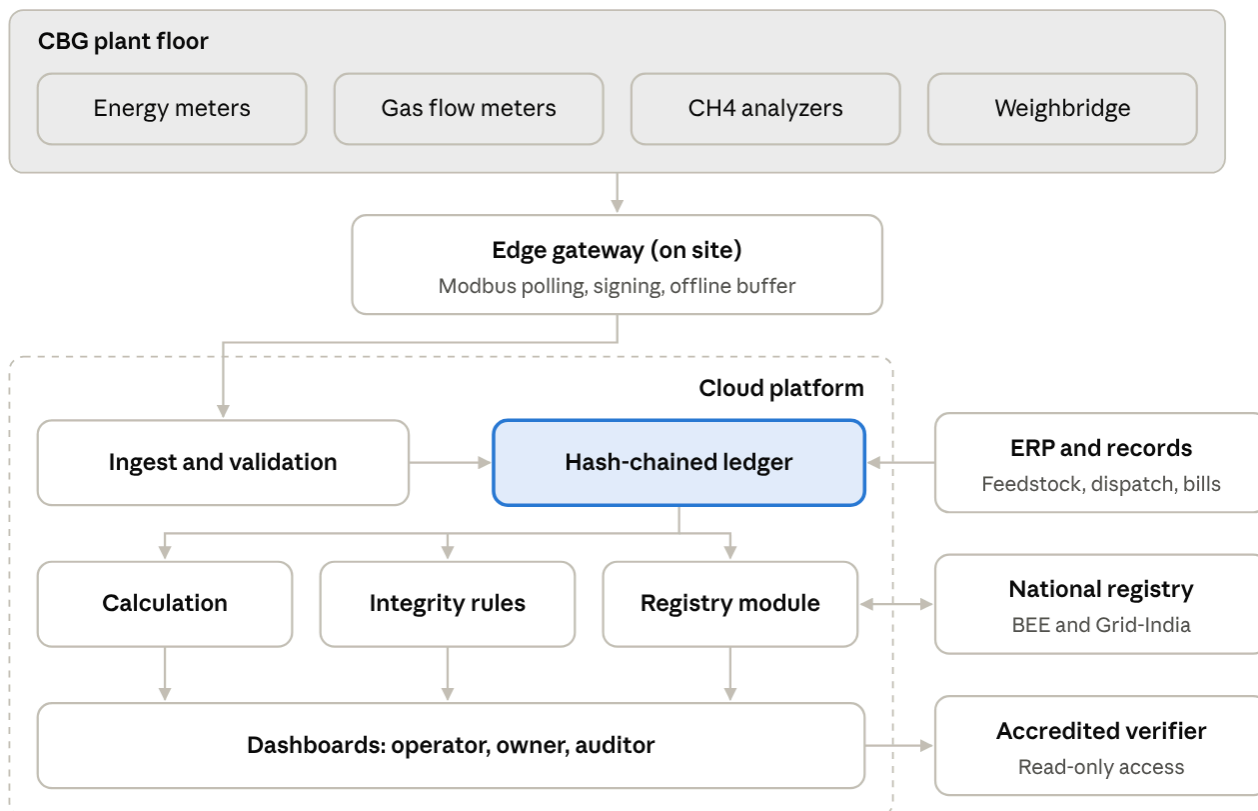
Key assumptions.

1. Credits follow an approved CCTS offset methodology for methane avoidance and fossil fuel displacement; the exact methodology must be confirmed before credits are claimed.
2. CBG counted for displacement credits must meet IS 16087; the 2016 edition sets a 90% methane minimum and a draft revision proposes 95% ([BIS draft](#)).
3. Grid emission factor comes from the current CEA CO2 Baseline Database; methane global warming potential follows the methodology.
4. Sensors are calibrated on a fixed schedule, and calibration records are stored in the ledger.

System architecture

Data moves in one direction: plant sensors to the edge gateway, to the cloud ledger, then to the calculation engine, registry and dashboards. Feedstock receipts and CBG dispatch records join in the cloud so energy and emissions can be divided by saleable output.

Every reading is validated and chained before it can become a credit



system architecture · plant, edge, cloud and external systems

All calculation, registry and dashboard services read from the hash-chained ledger; production records come in from ERP or Tally, and credit serials are reconciled with the national registry.

Sensors per plant (pilot: 5 TPD CBG plant).

Point	Sensor	Measures	Feeds which part of the credit
Feedstock gate	Weighbridge with digital output	Tonnes received, by feedstock type	Baseline: methane avoided
Raw biogas line	Thermal mass or ultrasonic gas flow meter	Nm3 per hour	Methane balance check

Point	Sensor	Measures	Feeds which part of the credit
Raw biogas and product gas	Online gas analyzer	CH ₄ , CO ₂ , H ₂ S, O ₂ (%)	Methane balance; product quality check
CBG dispatch	Cascade mass flow meter or dispenser totaliser	kg CBG sold	Baseline: fossil fuel displaced
Upgrading off-gas	Flow meter plus CH ₄ sensor	Methane in off-gas	Project emissions: methane slip
Digester domes, valves, compressor area	Fixed methane leak detectors	CH ₄ concentration in air	Project emissions: fugitive leaks
Flare (if present)	Flow meter and flame detector	Gas flared, flare operating	Project emissions: unburned methane
Main incomer and major motors	Energy meters, class 0.5S and clamp-on CT	kWh	Project emissions: electricity
Diesel generator and site vehicles	Fuel flow meter or fuel log	Litres	Project emissions: diesel
Digestate output	Weighbridge or volume log	Tonnes of digestate	Project emissions from digestate handling, if the methodology requires it

Edge layer. A low-cost industrial gateway polls meters and analyzers over Modbus RTU or TCP every 10 to 60 seconds, timestamps readings with GPS or NTP time, signs each batch, buffers up to 30 days offline and sends over MQTT on 4G or Wi-Fi.

Cloud layer. An ingest service validates readings and writes them to the hash-chained ledger. The calculation engine, integrity checks, registry module and dashboards all read from that ledger, never from raw unvalidated data.

Integration with plant systems.

- **Feedstock and dispatch records:** imported daily from Tally, spreadsheets or the plant's ERP; matched against weighbridge and dispatch meter readings.
- **SCADA or PLC (where present):** read-only Modbus or OPC UA connection for upgrader and compressor data; no control commands are sent.
- **Electricity bills:** uploaded monthly as a reconciliation check.

- **Registry and markets:** the registry module exports CCTS-aligned monitoring reports for the verifier and mirrors the national registry held by Grid Controller of India.

How the pipeline works, end to end

A truckload of feedstock becomes a verified credit in eight steps, and every step leaves a record in the ledger.

1. **Feedstock arrives.** The weighbridge records gross and tare weight; the operator tags the feedstock type (paddy straw, press mud, cattle dung, MSW organic fraction) on a tablet. The receipt is matched to the supplier's delivery note.
2. **Gas is produced and measured.** Flow meters and the gas analyzer record raw biogas volume and composition every minute.
3. **Gas is upgraded and compressed.** Off-gas flow and methane content are measured to quantify slip; product gas composition is checked against IS 16087.
4. **CBG is dispatched.** The dispatch meter records kg sold, matched to the invoice and buyer.
5. **Losses and energy are tracked.** Leak detectors, flare meters, energy meters and fuel logs record everything the project emits.
6. **Readings are validated and chained.** The edge gateway signs each batch; the cloud checks it against the validation rules and appends it to the hash-chained ledger.
7. **The period closes and credits are calculated.** At the end of each monitoring period (monthly internally, quarterly or yearly for issuance), the engine applies the methodology, resolves or excludes flags, and produces a monitoring report stamped with the period's Merkle root.
8. **Verification and issuance.** An accredited verifier reviews the report and the ledger remotely; BEE issues Carbon Credit Certificates into the national registry, and the registry module links each certificate to its evidence.

What the system checks at each step. The weighbridge reading must match the delivery note. Methane in must equal methane out plus losses, within meter uncertainty. Dispatched kg must match invoices. Product gas must meet the quality limit to count. Any mismatch flags the period before a single credit is calculated.

Credit calculation rules

Credits equal the emissions that would have happened without the plant, minus the emissions the plant itself causes. Every term comes from validated readings and fixed factors, so a verifier can reproduce the number by hand.

$$ER = BE_{CH_4} + BE_{fossil} - PE - LE$$

Baseline emissions (what the plant prevents).

- **Methane avoided, BE_{CH4}:** the methane the feedstock would have released if dumped, left to rot or burned in the open. It is calculated from weighed tonnes of each feedstock type using the methodology's model and default factors (for example, degradable organic carbon and methane correction factor for dumped waste).
- **Fossil fuel displaced, BE_{fossil}:** the CO₂ from the fossil fuel the CBG replaces.

$$BE_{fossil} = M_{CBG} \times NCV_{CBG} \times EF_{fossil}$$

where M_{CBG} is on-spec CBG sold in kg, NCV is its net calorific value, and EF is the emission factor of the displaced fuel (CNG or diesel, as the methodology specifies).

Project emissions (what the plant emits), PE.

$$PE = E_{grid} \times EF_{grid} + V_{diesel} \times EF_{diesel} + (M_{slip} + M_{leak} + M_{flare}) \times GWP_{CH_4} + PE_{digestate}$$

Here E is grid electricity in kWh, V is diesel in litres, M_{slip} is methane in upgrading off-gas, M_{leak} is fugitive methane, M_{flare} is unburned methane from flaring, and PE_{digestate} covers digestate storage or handling if the methodology requires it.

Leakage, LE: emissions shifted outside the plant, such as feedstock diverted from another use, as defined by the methodology.

Calculation rules.

ID	Rule
C1	Feedstock earns methane-avoidance credit only if it is weighed at the gate, its type recorded, and its baseline fate (dumping, burning, landfill) documented.
C2	Only CBG that is dispatched, invoiced and meets the quality limit counts toward displacement credits; flared, vented or off-spec gas is excluded.
C3	Methane slip and leaks are always measured, never assumed zero; if a leak detector is offline, the highest leak rate from the past 30 days is applied.

ID	Rule
C4	All emission factors, calorific values and GWP values are stored in configuration with source and version: CEA database for grid electricity, IPCC defaults or methodology values for the rest.
C5	Where two valid estimates of a term differ, the one giving fewer credits is used.
C6	Methane avoided cannot exceed the methane actually captured and accounted for in the plant's methane balance, unless the methodology explicitly allows it.
C7	Credits are calculated per monitoring period and rounded down to whole tonnes of CO ₂ e; fractions carry forward.
C8	A period with unresolved integrity flags at close earns no credits for the affected data.
C9	Factors are updated only at the start of a new period; past periods are never recalculated with new factors unless the verifier requires it.
C10	Final issuance follows the approved CCTS methodology; where it differs from these rules, the methodology wins.

Data integrity rules

No reading reaches a credit unless it passes validation, and every doubt is resolved toward fewer credits (the principle of conservativeness).

Each reading carries a status: `valid`, `flagged`, `substituted` or `excluded`. Only `valid` and `substituted` readings enter calculations. The dashboard always shows confirmed credits and credits on hold separately.

Validation and anomaly rules.

ID	Check	Triggers when	Result
V1	Completeness	A required field is missing or the device signature fails	Rejected at ingest

ID	Check	Triggers when	Result
V2	Physical range	Value outside physical limits (negative flow, methane above 100%)	Flagged
V3	Missing data	No reading from a sensor for more than 15 minutes while the plant runs	Gap flagged
V4	Energy balance	Sub-meters differ from the main incomer by more than 5%	Period flagged
V5	Bill reconciliation	Monthly metered kWh differs from the utility bill by more than 3%	Month flagged
V6	Methane balance	Methane in CBG dispatched plus off-gas plus flared exceeds methane in raw biogas by more than meter uncertainty	Period flagged
V7	Yield range	Biogas per tonne of feedstock outside the plausible range for that feedstock type	Period flagged
V8	Dispatch match	CBG dispatch meter differs from invoiced kg by more than 2%	Period flagged
V9	Quality	Product methane below the IS 16087 or contract limit	Gas excluded from output
V10	Clock integrity	Timestamps out of order or drift above 60 seconds	Batch flagged

Resolution rules.

ID	Rule
R1	A flagged reading leaves the credit total immediately and stays out until resolved.
R2	A confirmed real event, backed by independent evidence such as a weighbridge slip, invoice or lab gas analysis, is restored to <code>valid</code> .
R3	A sensor fault is filled from a backup in this order: a redundant meter (for example, dispatch totaliser for CBG flow), then invoices or bills pro-rata, then the most conservative comparable value.
R4	With no reliable backup, the gap earns zero credits.

ID	Rule
R5	Suspected tampering is excluded permanently, logged as an incident and reported to the verifier.
R6	Flags unresolved at the end of the reporting period are treated as excluded .

Ledger rules.

ID	Rule
L1	Every reading is stored with the SHA-256 hash of its content plus the previous record's hash.
L2	Records are never edited or deleted. A correction is a new record referencing the original, with the reason and approver.
L3	The chain is verified hourly and on every report export; a break raises a critical alert and freezes issuance for the affected period.
L4	Each period's records are summarised into a Merkle root, stamped onto the monitoring report and every credit issued from it.
L5	Auditors have read-only access to the full chain and can re-run verification themselves.

Role of machine learning. Rules make every decision on credits. Once months of real data exist, ML can assist by predicting expected gas output from feedstock as a cross-check, spotting subtle multi-sensor anomalies and suggesting the likely cause of a flag. ML never sets a substituted value or approves a reading.

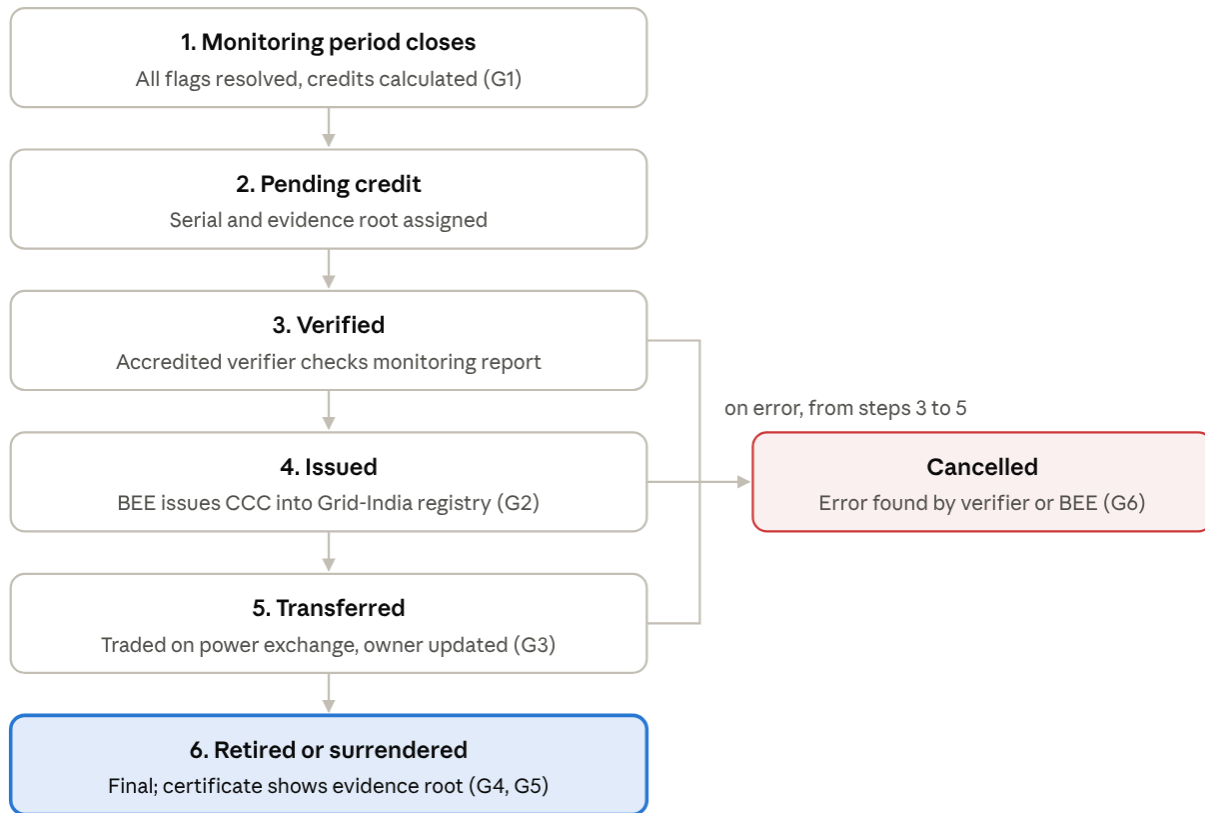
Carbon registry

The registry module tracks every credit from verified reduction to final retirement, and links each credit back to the exact sensor records that produced it. It is a project-level sub-registry: official certificates are issued only by the Bureau of Energy Efficiency (BEE) into the national registry, and the module mirrors and reconciles with that record.

India's institutions. Under the CCTS, BEE administers the scheme and issues Carbon Credit Certificates (CCCs), Grid Controller of India operates the registry, and the Central Electricity Regulatory Commission (CERC) regulates trading ([Tata Power](#)). CERC notified

its regulations on purchase and sale of CCCs on 27 February 2026, and BEE has published its offset procedure and list of accredited verifiers ([IMPRI](#)). Certificates trade on the power exchanges ([CERC draft regulations](#)). One CCC equals one tonne of CO₂e.

A credit is final only after retirement, and every step is logged



credit lifecycle · six steps and the cancellation path

A holder can also retire an issued credit directly without a trade; every arrow is a registry event written to the hash chain (rule G8).

Credit record. Every credit carries these fields:

Field	Example	Purpose
Serial number	DV-GHY01-2026Q3-000142	Unique ID: project, plant, vintage, sequence
Vintage period	2026-07-01 to 2026-09-30	When the reduction happened
Methodology	CCTS offset methodology ID and version	Rulebook used
Plant unit	GHY01, Dross Vault Guwahati CBG plant	Which plant earned it

Field	Example	Purpose
Evidence root	Merkle root of the period's ledger records	Links the credit to its sensor data
Monitoring report hash	SHA-256 of the verified report	Links the credit to the verified document
Verifier	Accredited Carbon Verification Agency name	Who checked it
National registry ID	CCC serial from the Grid Controller of India registry	Reconciliation with the official record
Status	Pending, issued, transferred, retired, surrendered, cancelled	Lifecycle state
Owner account	Current holder	Who holds it now

Registry rules.

ID	Rule
G1	Credits are created in pending only after the monitoring period closes and all flags are resolved.
G2	A pending credit becomes issued only when the matching CCC appears in the national registry; the national serial is recorded against it.
G3	Each credit has one owner at a time. Transfers are recorded as events; balances are derived from events, never edited.
G4	retired and surrendered are final. A final credit can never be transferred again, which prevents double counting.
G5	Every retirement records the beneficiary and purpose, and produces a retirement certificate showing the evidence root.
G6	If a verifier or BEE later finds an error, affected credits move to cancelled with a reason; nothing is deleted.
G7	A buffer of 5% of each issuance is held back in a portfolio reserve to cover later corrections.

ID	Rule
G8	Registry events (issue, transfer, retire, cancel) are written to the same hash chain as sensor data.
G9	Order matching and price discovery happen off-chain on the power exchanges; only lifecycle events are recorded in the chain, following the split proposed in recent research (Taylor & Francis, 2026).
G10	The record shows each credit's current location and adjustment status publicly, as recommended for interoperable registries (CFI and Max Planck).

What makes this different: evidence-linked credits. Most registries record that a credit exists. Here, anyone holding a credit can follow its evidence root back to the exact meter readings, flags and corrections behind it, and re-verify the hash chain independently. Recent work on blockchain registries stresses unique serials and full lifecycle tracking to prevent double counting ([npj Climate Action, 2026](#)); this design adds the link to raw measurements without requiring a public blockchain.

Portfolio aggregation for smaller plants

Larger CBG plants can register their own projects, but 2 to 5 TPD plants benefit from pooling into one portfolio project so that registration and verification costs are shared. Each plant is still paid for its own measured reductions.

Roles.

Role	Who	Responsibility
Aggregator and project proponent	Dross Vault, a CBG developer or an oil company offtaker	Registers the portfolio, signs plant agreements, holds the project registry account
Participating plant	Individual CBG plant	Hosts sensors, follows data rules, receives its share of credit revenue
Platform operator	Dross Vault	Runs sensors, ledger, calculations and registry module

Role	Who	Responsibility
Verifier	BEE-accredited verification agency	Verifies the portfolio monitoring report with remote ledger access
Administrator and registry	BEE and Grid Controller of India	Issue and hold the official certificates

Allocation rule. Each plant's credits are its own verified reduction, after the shared buffer and fee:

$$\text{Credits}_{\text{plant}} = ER_{\text{plant}} \times (1 - b) \times (1 - f)$$

where b is the portfolio buffer (5%) and f is the platform fee share (proposed 15%).

Aggregation rules.

ID	Rule
A1	Each plant has its own baseline; plants are never averaged together for crediting.
A2	A plant joins only after 4 weeks of clean baseline data and a signed data-sharing agreement.
A3	A plant failing integrity rules loses credits for that period only; other plants are unaffected.
A4	Plants can join in later vintages without re-registering the portfolio, if they use the same methodology.
A5	A plant can exit at the end of any vintage; credits already issued stay with it.
A6	Revenue is paid to plants within 30 days of each sale, with a statement of credits, price and fee.
A7	Each plant's data is visible only to that plant, the aggregator and the verifier; portfolio dashboards show anonymised rankings.

Data model

Twelve tables hold everything from raw readings to retired credits; readings, corrections and registry events are append-only.

Table	Key fields	Notes
project	project_id, name, aggregator_id, methodology_id	One registered project or portfolio
plant	plant_id, project_id, name, capacity_tpd, feedstock_types, joined_on, status	One CBG plant
device	device_id, plant_id, type, meter_class, calibration_due, public_key	Signing key verifies readings
reading	reading_id, device_id, ts, quantity, value, unit, status, hash, prev_hash	Append-only, hash-chained
feedstock_receipt	receipt_id, plant_id, ts, feedstock_type, tonnes, source_location, weighbridge_ref	Basis for methane-avoidance credits
dispatch	dispatch_id, plant_id, ts, kg_cbg, ch4_pct, buyer, invoice_ref	Basis for displacement credits
flag	flag_id, reading_id or period_id, rule_id, opened_at, resolution, resolved_by	Links to rules V1 to V10
correction	correction_id, original_reading_id, new_value, reason, approver, hash	Never overwrites the original
emission_factor	factor_id, source, value, unit, version, valid_from	CEA, IPCC and methodology factors
period	period_id, plant_id, start, end, be_ch4, be_fossil, pe_total, ch4_slip_kg, ch4_leak_kg, er_tco2e, flags_resolved, merkle_root	One reporting period per plant
credit	serial, period_id, plant_id, vintage, methodology_id, evidence_root, report_hash, national_ccc_id, status, owner_account	One row per tonne

Table	Key fields	Notes
registry_event	event_id, serial, type, from_account, to_account, ts, reason, hash, prev_hash	Issue, transfer, retire, surrender, cancel

Balances, totals and dashboards are always computed from these tables and never stored as editable figures.

UX dashboards

Three screens serve three users, each showing only what that person acts on.

1. Plant operator (mobile first, local language).

- Data health: which sensors are online, last reading time, any gaps.
- Today's feedstock received by type, gas produced, CBG dispatched.
- Methane balance for the day: in, out, losses, with a green or red status.
- Active flags and leak alarms, with the action needed to resolve each.
- Calibration due dates for each meter.

2. Plant owner or portfolio manager (desktop).

- Credits this period: confirmed, on hold, and the main reasons for holds.
- Breakdown of credits by source: methane avoided, fossil fuel displaced, minus project emissions.
- Credits pipeline: pending, verified, issued, sold, retired.
- Open flags by plant and rule, with ageing.
- Credit revenue statement per plant.

3. Auditor and verifier (read-only).

- Period selector with every term of the credit calculation and the factors used.
- Full reading table with hashes; Verify chain button that re-runs the check.
- Flags and corrections log with reasons and approvers.
- Credit lookup by serial: evidence root, report hash, national registry ID and full event history.
- One-click export of the monitoring report in the format the methodology requires.

Worked example: one year at a 5 TPD plant

For an illustrative 5 TPD plant running 330 days, displacement credits less project emissions come to about 1,770 tCO₂e a year before methane-avoidance credits are added. Every input below is an assumption to be replaced with measured data and methodology values.

Term	Inputs (assumed)	tCO ₂ e per year
Fossil fuel displaced	1,650 t CBG × 50 MJ per kg × 56.1 tCO ₂ per TJ (natural gas)	+4,628
Grid electricity	3,300 MWh × 0.71 tCO ₂ per MWh	-2,343
Methane slip and leaks	16.5 t CH ₄ (1% of output) × GWP 28	-462
Diesel	20,000 litres × 2.68 kgCO ₂ per litre	-54
Subtotal before methane avoided		+1,769
Methane avoided	Weighed feedstock by type × methodology model	Added on top; depends on feedstock and its baseline fate

Why measuring methane loss matters. Each 1% of output lost as methane is about 16.5 t of CH₄, or 462 tCO₂e a year at GWP 28: more than a quarter of the subtotal above. A plant that assumes zero slip overstates its credits; a plant that cannot prove low slip may be assigned a conservative default that costs it credits. Continuous measurement protects both the buyer and the plant.

Manual MRV compared with dMRV.

Aspect	Manual MRV	Dross Vault dMRV
Feedstock data	Handwritten weighbridge slips	Logged automatically, tagged by type, matched to delivery notes
Gas and CBG data	Periodic readings, invoices	Continuous flow and composition, matched to invoices
Methane loss	Often assumed or a default	Measured continuously

Aspect	Manual MRV	Dross Vault dMRV
Error and fraud detection	Found at audit, if at all	Flagged within minutes by validation rules
Tamper evidence	None	Hash chain; any edit is detected
Verification	Site visit, paper reconciliation	Remote read-only access, re-runnable checks
Report preparation	Weeks of compilation	Generated when the period closes
Buyer confidence	Depends on trust in the developer	Each credit traceable to its sensor records

Deployment and business model

A full kit costs about INR 12.5 lakh per plant, less where the plant already has gas analyzers or flow meters. The plant pays it back through credits that are issued faster, verified more cheaply and trusted more by buyers. All prices are estimates to be confirmed with vendor quotes.

Installation cost per plant (estimate).

Item	Quantity	Cost (INR)
Gas flow meters (raw biogas, off-gas, product)	3	4,50,000
Online gas analyzer (CH ₄ , CO ₂ , H ₂ S, O ₂)	1	4,00,000
Fixed methane leak detectors	6	1,50,000
Energy meters (main incomer and major motors)	1 + 8	89,000
Digester temperature, pressure and pH probes	1 set	50,000
Weighbridge and dispatch meter integration	1	30,000
Edge gateway with 4G	1	30,000
Installation and commissioning	1	50,000
Total		12,49,000

Credit revenue sensitivity. Carbon Credit Certificate prices are not yet established; first trading on the power exchanges was expected around October 2026 ([Orennow](#)). Using only the 1,769 tCO₂e subtotal from the worked example, with methane-avoidance credits excluded:

Assumed price per credit (INR)	Annual credit revenue (INR)	Time to recover year-1 cost of INR 17.5 lakh
500	8.8 lakh	About 24 months
1,000	17.7 lakh	About 12 months
2,000	35.4 lakh	About 6 months

Year-1 cost is hardware plus an assumed INR 5 lakh annual subscription. Methane-avoidance credits shorten these periods further.

Other value to the plant. Lower verification cost through remote audits; faster issuance because reports are ready when a period closes; fewer disputes with CBG buyers because quality and volume are independently logged; and early warning of leaks and equipment faults from the same sensor data.

Revenue models.

1. **Purchase plus subscription:** plant buys the kit and pays a yearly fee for the platform, reports and registry.
2. **Zero capex plus carbon share:** Dross Vault installs at no upfront cost and takes a share of credit revenue (proposed 25%) for the project's crediting period.
3. **Carbon fee:** 15% of credit revenue on top of the subscription for portfolio projects.

Market size (CBG plants only, at INR 12.5 lakh hardware and INR 5 lakh a year subscription).

Segment	Plants	One-time hardware (INR)	Recurring per year (INR)
Commissioned today	132	~16.5 crore	~6.6 crore
Holding SATAT letters of intent	~1,100	~137.5 crore	~55 crore
SATAT 5,000-plant target	5,000	~625 crore	~250 crore

Go-to-market. Start with the Dross Vault Guwahati plant as the reference site. Then sell to CBG developers running several plants, through EPC and upgrader vendors who can bundle the kit with new builds, and through the Indian Biogas Association. Oil marketing companies buying CBG under SATAT are natural partners, because verified quality and volume data reduce their own checks.

Scale-up pathway.

1. **Pilot:** Guwahati plant; run dMRV alongside manual records for one full monitoring period; first verified report.
2. **Early customers:** 5 to 10 plants with 2 or 3 developers; portfolio registry for smaller plants.
3. **Scale:** 50 or more plants; kit bundled into new SATAT builds by EPC partners.
4. **Platform:** registry module offered as a service to other dMRV providers and aggregators; extension to MSW and press-mud bio-CNG plants.

Software prototype plan

The prototype is a laptop simulation of a 5 TPD CBG plant (plus two smaller plants for the portfolio view) that runs the full pipeline live, from simulated meters to a retired credit. No hardware is required.

Module	Function	Tool
<code>simulator.py</code>	Simulates feedstock receipts, biogas flow, methane %, electricity by equipment and CBG dispatch, with fault switches: inflate dispatch, methane leak, compressor idling, dead sensor	Python
<code>api.py</code>	Ingests and validates readings (rule V1)	FastAPI
<code>ledger.py</code>	Hash chain, Merkle root per period, chain verification (rules L1 to L5)	SQLite, SHA-256
<code>rules.py</code>	Validation, anomaly and resolution rules V2 to V10, R1 to R6	Python
<code>engine.py</code>	Baseline emissions, project emissions, leakage, methane balance and ER (rules C1 to C10)	Python, pandas
<code>registry.py</code>	Credit creation, serials, transfer, retirement, cancellation (rules G1 to G10) and plant allocation (rules A1 to A7)	Python
<code>dashboard.py</code>	Operator, owner and auditor screens	Streamlit
<code>report.py</code>	Monitoring report and retirement certificate PDFs	ReportLab

Demo script (5 minutes).

1. Plant runs normally: feedstock, gas, dispatch and project emissions stream in; credits for the period build up live, split into methane avoided, fossil displaced and project emissions.

2. Inflate-dispatch fault: dispatch match rule V8 and methane balance rule V6 flag it; credits move to on hold, confirmed credits unchanged.
3. Methane leak fault: leak alert fires with location; project emissions rise and credits fall accordingly.
4. Dead sensor: gap flagged; backup source substituted conservatively and marked substituted .
5. Edit an old reading in the database: chain status turns red and names the record; issuance freezes.
6. Period closes: monitoring report generated; credits created with serials and evidence roots; one is transferred and retired, producing a certificate that traces back to its sensor data.

Risks and open questions

The biggest risks are methodology availability under the CCTS and verifier acceptance of digital data; both are reduced by running dMRV alongside manual records during the pilot.

Risk	Impact	Mitigation
No approved CCTS methodology yet fits CBG methane avoidance and displacement	Credits delayed	Engage BEE early; design rules to map onto standard CDM-style methodologies; sell into voluntary standards meanwhile
Verifiers unwilling to rely on remote data	Higher verification cost	Read-only auditor access, re-runnable chain checks, parallel manual logs in the pilot
Gas meter and analyzer drift	Wrong credits	Calibration schedule, methane balance (V6) and dispatch match (V8) rules, redundant metering on key flows
Feedstock type misreported to claim higher avoidance credits	Overstated credits	Delivery-note matching, yield range check (V7), periodic feedstock sampling
Sensor or gateway failure at remote sites	Data gaps	Offline buffering, conservative substitution rules R3 and R4, spare-parts kit
Low or volatile CCC prices	Weaker revenue	Subscription value from faster verification and operational alerts, not only credit sales

Open questions.

- ☐ Which approved CCTS offset methodology applies to CBG plants, and does it require measured methane slip?
- ☐ Which accredited verification agency will review the Guwahati pilot?
- ☐ Which sensors does the Guwahati plant already have, so the kit cost can be reduced?
- ☐ Which voluntary standard, if any, should be used until a CCTS methodology is approved?

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